

FORBIOM-ON

(a dynamic, spatial model for ontario forest market)

Technical Model Documentation

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DRAFT

Contents

1	Overview	3
2	Existing models for wood markets	4
3	Sets and Indices	4
4	Data and Parameters	5
4.1	Forest Yield and Biological Growth	5
4.2	Economic Flow Parameters	5
5	Model Variables	6
6	Equations	6
6.1	Biological and Harvesting Constraints	6
6.2	Manufacturing and Logistic constraints	7
6.3	Objective Function and Carbon Accounting	7
7	Results (from a preliminary counterfactual run)	8

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Abstract

FORBIOM-ON is a spatial, multi-commodity model for wood and wood-based products with a focus on forest regions in Ontario. It is designed to analyze the impacts of market developments and policy changes on forest management, wood processing industries, and regional wood product markets. The model represents the interactions between forest resources, industrial production, and inter-regional trade in major wood products, including lumber, pulpwood, and engineered wood.

The framework integrates forest growth dynamics, harvesting decisions, manufacturing processes, and transport flows within a consistent economic structure. Implemented in GAMS and formulated as a nonlinear programming (NLP) problem, the model allows the analysis of how changes in market conditions, logistics constraints, industrial capacity, and carbon pricing influence harvest levels, production patterns, regional prices, and trade flows across the forest value chain. Preliminary simulations indicate that the model is able to capture regional developments in the Ontario forest sector and to evaluate the implications of policy and market changes for supply, demand, industrial production, and carbon outcomes.

1 Overview

FORBIOM-ON is designed to analyze the economic and environmental impacts of provincial policy shifts and market fluctuations on the forest industry and its related value chains in Ontario. It is a regionalized economic simulation model that evaluates the trade-offs between biological forest preservation, industrial production, and carbon sequestration goals. The project is a research initiative within the University of Toronto’s Faculty of Forestry, focused on the advancement of the provincial bioeconomy and mass timber sector.

The model is currently operational for a ten-year planning horizon (2020–2030) and integrates regional biophysical yield data with industrial capacity and logistics constraints across Northern and Southern Ontario.

Generally, the model belongs to the class of spatial equilibrium and multi-commodity models. Its core characteristics are:

- **Recursive-dynamic with a yearly resolution**, where forest stock levels and available harvests in the current period are functionally dependent on the biological increment and extraction rates of previous years.
- **Distinction between seven tree species and site class** (Spruce, Jack Pine, White Birch, Tolerant Hardwoods, White Pine, Red Pine, and Aspen) and three Site Classes (SC1, SC2, SC3).
- **Supply-side differentiation based on Plonski Yield Tables**, which drive species-specific Gross Merchantable Volume (m^3/ha) as a function of forest age and dominant site class.
- **Manufacturing hierarchy disaggregated by layers**: Primary (Lumber, Pulpwood) and Secondary (CLT, Glulam), where secondary production is physically constrained by the mass-balance availability of regional lumber inputs.
- **Regionalized at the Forest Management Unit level** for six key Ontario regions: Trout Lake, Abitibi, Algoma, Kenora, Nipissing, and Algonquin.
- **Bi-lateral trade flows based on the Spatial Equilibrium concept**, which calculates optimal inter-regional shipments by solving for spatial arbitrage conditions subject to transport arc capacities and freight costs.

- **Growth engine driven by Current Annual Increment (CAI)**, where annual roundwood accumulation is interpolated from yield-age brackets and applied to calculated regional forest areas.
- **Integrated carbon accounting system** that evaluates the net balance between process-based manufacturing emissions, freight-related transport emissions, and long-term carbon storage factors within the harvested wood product (HWP) pool.
- **Based on data from the Ontario Ministry of Natural Resources and Forestry (MNR)**, regional species compositions, and industrial mill capacity benchmarks.

The model and related data transformations are encoded in GAMS. It is defined as a Non-Linear Program (NLP) and solved using high-performance solvers (e.g., CONOPT or SNOPT). Data exploitation and result processing are handled through GDX-to-CSV translation for integrated reporting of regional revenue, manufacturing tier metrics, and carbon value chains.

2 Existing models for wood markets

Forbiom-ON belongs to the group of recursive-dynamic partial equilibrium models for the forest sector. The history of these models goes back to the Global Trade Model (GTM), which was the first to study how wood products move between countries (Buongiorno and Gilles, 1983). Around the same time, the PAPHYRUS model was created to look at the pulp and paper industry in North America (Buongiorno and Gilles, 1987). These early models created the foundation needed to combine forest growth with market supply and demand.

The most well-known models in this group are the EFI-GTM (Kallio et al., 2004) and the GFPM (Tomberlin and Buongiorno, 1998). The EFI-GTM is used to study wood markets mostly in Europe. The Global Forest Products Model (GFPM) is similar but covers the whole world, dividing it into 180 different regions to see how trade flow on a global scale.

Because global models are too broad to show local policy impacts, researchers have built models for specific countries and regions. Examples include the Norwegian Trade Model (Trømborg and Sjølie, 2011), the Finnish Forest Sector Model (Ronnala, 1995), and the French Forest Sector Model (Cauria et al., 2010). These models use high-resolution local data to provide better policy insights for specific jurisdictions.

The Forbiom-ON contributes to this field by providing a comprehensive **vertically integrated framework** for the Ontario forest sector. While traditional models often treat the forest and the manufacturing sector as separate units, Forbiom-ON creates a direct, physical link between the biological growth of specific tree species and the production limits of advanced wood industries. By accounting for the strict industrial dependencies between primary sawmilling and secondary manufacturing, the model ensures that the expansion of the mass timber sector is balanced against the actual sustainable harvest limits of the province. This approach allows for a realistic evaluation of how regional forest management policies impact the entire value chain. This makes it a practical tool for assessing if new wood policies are both environmentally sustainable and industrially possible.

3 Sets and Indices

The model dimensions are defined by the following sets:

$r, rr \in \mathcal{R}$ Regions: {TroutLake, Abitibi, Algoma, Kenora, Nipissing, Algonquin}

$p \in \mathcal{P}$ Products: {CLT, Glulam, Lumber, Pulpwood}

$y \in \mathcal{Y}$ Years: {2020, ..., 2030}

$tree \in \mathcal{T}$ Tree Species: {Spruce, JackPine, WhiteBirch, TolerantHdwds, WhitePine, RedPine, Aspen}

$s \in \mathcal{S}$ Site Classes: {SC1, SC2, SC3}

$age \in \mathcal{A}$ Forest Age Cohorts: {0, ..., 150}

$m \in \mathcal{M}$ Manufacturing Layers: {Primary, Secondary}

4 Data and Parameters

4.1 Forest Yield and Biological Growth

Instead of assumed generic growth rates, the model derives Current Annual Increment (CAI) dynamically using Plonski yield tables (Gross Merchantable Volume in m³/ha).

YieldData(tree, s, age) Merchantable volume per hectare based on species, site class, and age.

speciesMix(r, tree) Fractional composition of forest species within a given region.

initialStockRegion(r) The starting inventory of roundwood (in million m³).

regionArea(r) Implied forest area (in ha), back-calculated by dividing the total regional stock by the weighted average yield of the region's initial age cohort.

CAI_Region(r, y) Annual roundwood growth (in million m³/year), calculated via a 10-year forward bracket of the Plonski tables applied to the calculated *regionArea*.

4.2 Economic Flow Parameters

basePrice(p) Starting market price in 2020 for product p (in \$/m³).

basePrice_y(p, y) Exogenous price path, inflating *basePrice(p)* at 3% annually.

processingCost_y(p, y) Total processing cost per unit, composed of a tier base cost (Primary/Secondary) plus a product-specific markup, decaying at 2% annually due to presumed efficiency gains.

transportCost(r, rr, p) Inter-regional logistics cost (in \$/m³).

$\eta(p)$ Primary yield factor (e.g., 0.90 for Lumber, meaning 1 m³ of roundwood yields 0.90 m³ of lumber).

$\rho(p)$ Lumber input requirement for secondary products (e.g., 1.20 m³ of lumber per m³ of CLT).

5 Model Variables

All volume variables are continuously defined in **Millions of cubic meters**, and monetary variables in **Millions of dollars**.

stockRW(r, y) Standing roundwood inventory.

harvestRW(r, y) Total roundwood harvested.

production(r, p, y) Output volume of final products.

sales(r, p, y) Final volume sold to fulfill regional demand.

exports(r, rr, p, y) Volume of product transported between regions.

price(r, p, y) Fixed variable bounded to the *basePrice_y* path.

Zprofit Net economic value before carbon penalties.

Zemission Aggregate carbon emissions across all activities (MtCO₂ eq).

carbonCost Total systemic carbon tax cost.

obj The final maximization objective: $Zprofit - carbonCost$.

6 Equations

6.1 Biological and Harvesting Constraints

Stock Balance: Forest stock is strictly roundwood. Stock in any year is the previous year's stock plus calculated Plonski growth, minus the harvested volume. The growth calculation is based on the Plonski Yield Tables (Plonski, 1971). These tables provide standard growth rates for Ontario's tree species based on their age and site quality. The specific data used for these growth functions is sourced from the [Normal Yield Tables for Ontario](#) (Lakehead University, 2007). This ensures the forest inventory in the model grows at a realistic, biological rate.

The growth component is represented by $CAI_Region_{r,y}$, which stands for the Current Annual Increment. This is the total volume of wood a specific region (r) gains in a given year (y) through natural growth. In the model, this is not a fixed number; it is a weighted average based on the **Species Mix** (e.g., % Spruce vs. % Jack Pine) and the **Age** progression.

Because different trees grow at different speeds as they get older, the model calculates the growth by taking the difference between two age points on the Plonski curve and dividing by 10 to get a yearly rate. This rate is then multiplied by the calculated forest area for each region:

$$CAI_Region_{r,y} = regionArea_r \times \sum_{tree} \left(speciesMix_{r,tree} \times \frac{Yield_{tree,s,age+10} - Yield_{tree,s,age}}{10} \right) \quad (1)$$

The stock balance is then maintained using the following equation:

$$stockRW_{r,y} = \begin{cases} initStockRW_r - harvestRW_{r,y} & \text{if } y = 2020 \\ stockRW_{r,y-1} + CAI_Region_{r,y} - harvestRW_{r,y} & \text{if } y > 2020 \end{cases} \quad (2)$$

Harvest Sustainability: The annual harvest is capped at a maximum sustainable fraction ($maxHarvestRate$) of the total available biological mass.

$$harvestRW_{r,y} \leq \alpha_r \cdot \begin{cases} initStockRW_r & \text{if } y = 2020 \\ stockRW_{r,y-1} + CAI_Region_{r,y} & \text{if } y > 2020 \end{cases} \quad (3)$$

6.2 Manufacturing and Logistic constraints

Primary Processing (Roundwood Consumption): Harvested logs are the only way to produce first-level products, like lumber and pulpwood. The amount of product created is controlled by the yield efficiency (η). For example, if the efficiency is 0.90, then 1 m^3 of logs will produce 0.90 m^3 of lumber.

$$\frac{production_{r,Lumber,y}}{\eta_{Lumber}} + \frac{production_{r,Pulpwood,y}}{\eta_{Pulpwood}} \leq harvestRW_{r,y} \quad (4)$$

Nested Lumber Mass Balance: Lumber operates as both a final product and an intermediate input. The local lumber market must supply final *sales*, outward *exports*, and the physical material required (ρ) to manufacture CLT and Glulam.

$$production_{r,Lumber,y} + \sum_{rr \neq r} exports_{rr,r,Lumber,y} = sales_{r,Lumber,y} + \sum_{rr \neq r} exports_{r,rr,Lumber,y} + \rho_{CLT} \cdot production_{r,CLT,y} + \rho_{Glulam} \cdot production_{r,Glulam,y} \quad (5)$$

Secondary Product Mass Balance: For products with no downstream manufacturing dependencies (CLT, Glulam, Pulpwood):

$$production_{r,p,y} + \sum_{rr \neq r} exports_{rr,r,p,y} = sales_{r,p,y} + \sum_{rr \neq r} exports_{r,rr,p,y} \quad (6)$$

Capacity Constraints: Production and shipments are strictly bounded by physical infrastructure parameters.

$$production_{r,p,y} \leq millCap_{r,p,y} \quad (7)$$

$$exports_{r,rr,p,y} \leq arcCap_{r,rr,p,y} \quad \forall r \neq rr \quad (8)$$

$$sales_{r,p,y} \leq demandCap_{r,p,y} \quad (9)$$

6.3 Objective Function and Carbon Accounting

Carbon Accounting: Total emissions aggregate the process-based manufacturing factors and spatial freight emissions.

$$carbonEmission_{r,p,y} = E_{proc}(p) \cdot production_{r,p,y} + \sum_{rr \neq r} E_{trans}(r, rr, p) \cdot exports_{r,rr,p,y} \quad (10)$$

$$Z_{emission} = \sum_{r,p,y} carbonEmission_{r,p,y} \quad (11)$$

$$carbonCost = carbonPrice \cdot Z_{emission} \quad (12)$$

Economic Objective: The solver operates as a Non-Linear Program (NLP) maximizing the aggregate objective obj . Market prices are strictly bounded ($price.fx(r, p, y) = basePrice_y(p, y)$) to prevent solver manipulation.

$$\max obj = Zprofit - carbonCost \quad (13)$$

where $Zprofit$ is derived as:

$$\begin{aligned} Zprofit = & \sum_{r,p,y} (price_{r,p,y} \cdot sales_{r,p,y}) \\ & - \sum_{r,y} (harvestCostRW \cdot harvestRW_{r,y}) \\ & - \sum_{r,p,y} (processingCost_y_{p,y} \cdot production_{r,p,y}) \\ & - \sum_{r,rr,p,y} (transportCost_{r,rr,p} \cdot exports_{r,rr,p,y}) \end{aligned} \quad (14)$$

7 Results (from a preliminary counterfactual run)

(See attached)